

TECHNICAL UNIVERSITY OF CRETE

School of Electrical & Computer Engineering

Investigating the Applicability of Federated Dynamic Averaging in Federated Low-Rank Adaptation Architectures

Diploma Thesis Presentation

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Background Knowledge : Distributed Learning, FedAvg & Client Drift

- **Federated Learning (FL)**: Privacy-preserving collaborative optimization of foundation models without centralizing client data.
- **Federated Averaging (FedAvg)**:

$$w_{t+1} = w_t + \frac{1}{|S_t|} \sum_{k \in S_t} \Delta_t^{(k)}$$

Clients execute τ local SGD steps on private data $\mathcal{D}_k$.

- **Statistical Heterogeneity (Non-IID)**: Client distributions differ in label, quantity, or features. Modeled via **Dirichlet distribution** $p_k \sim \mathrm{Dir}(\alpha)$.
- **Client Drift**: Divergent local minima pull updates away from the true global optimum.

Dirichlet Heterogeneity (α)

- $\alpha \to 0$: Extreme label skew (clients hold single-class data).
- $\alpha \to \infty$: Uniform IID distribution.
- Key benchmark lever for evaluating drift robustness.

The Dual Communication Bottleneck

- **Spatial**: Model parameter volume (billions of weights).
- **Temporal**: Rigid synchronization cadence ($\tau = E$).

Background Knowledge : Low-Rank Adaptation (LoRA)

- Pre-trained weights $W_0 \in \mathbb{R}^{d \times k}$ are kept **frozen**.
- Updates factorized into low-rank matrices:

$$\Delta W = \frac{\alpha_{\text{lora}}}{r} BA, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \; B \in \mathbb{R}^{d_{\text{out}} \times r}$$

  where rank $r \ll \min(d_{\text{in}}, d_{\text{out}})$.
- Forward pass: $h = W_0 x + \frac{\alpha_{\text{lora}}}{r} BAx$.
- Trainable parameters reduced by $> 99\%$.
- **LoRA Adapter Hyperparameters**:
  - **Rank** $r = 4$: Low-rank bottleneck dimension.
  - **Scaling** $\alpha_{\text{lora}} = 8$: Scaling factor $\frac{\alpha_{\text{lora}}}{r} = 2.0$.
  - **Target Modules**: Query (W_q) and Value (W_v) projections.
  - **Initialization**: Gaussian for A, Zero for B (ensures $\Delta W = 0$ at start).

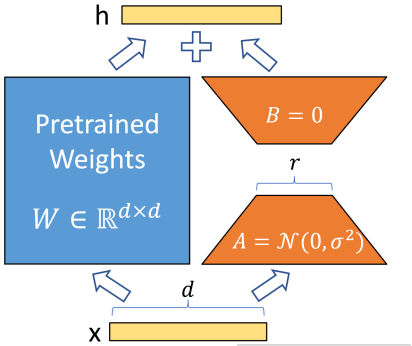


Figure: LoRA decomposition with frozen weights W_0.

Background Knowledge : The LoRA Aggregation Inconsistency Problem

Mathematical Non-Commutativity (Bilinear Distortion)

Independent server-side averaging of client low-rank factors fails to reconstruct the composite weight update:

$$\frac{1}{N}\sum_{i=1}^{N}(A_i B_i) \neq \left(\frac{1}{N}\sum_{i=1}^{N} A_i\right)\left(\frac{1}{N}\sum_{i=1}^{N} B_i\right)$$

- Direct FedAvg on (A_i, B_i) induces cross-client geometric interference.
- Destabilizes convergence under non-IID data distributions.
- **Prior Work Limitations**:
  - *Full-matrix expansion* ($\Delta W = AB$): Transmits full parameter matrices $\rightarrow$ destroys spatial savings.
  - *FFA-LoRA*: Permanently freezes down-projection A $\rightarrow$ sacrifices adapter expressivity.
  - *FLoRA*: Server-side SVD recalculation $\rightarrow$ high server compute overhead.

Alternating Coordinate Optimization Restores Aggregation Linearity

RoLoRA resolves the bilinear inconsistency by alternating the active trainable matrix while freezing the other globally across all clients:

- **Odd Rounds (Train B, Freeze A)**: $\frac{1}{N}\sum_{i=1}^{N}(B_iA) = \left(\frac{1}{N}\sum_{i=1}^{N}B_i\right)A = \bar{B}A$
- **Even Rounds (Train A, Freeze B)**: $\frac{1}{N}\sum_{i=1}^{N}(BA_i) = B\left(\frac{1}{N}\sum_{i=1}^{N}A_i\right) = B\bar{A}$
- **Core Limitation**: Enforces a rigid, static local epoch schedule (E), leaving temporal synchronization completely unoptimized.

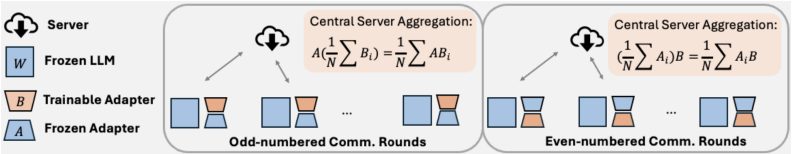


Figure: RoLoRA alternating framework across communication rounds.

Background Knowledge : FDA-Opt & Geometric Monitoring

Theologitis, Samoladas & Deligiannakis, 2025 – Dynamic Variance Halting

- **Geometric Monitoring Root**: Decomposes global threshold functions over distributed streams into local **safe zones** (Sharfman et al., 2006).
- **FDA-Opt Formulation** (Theologitis et al., 2025): Tracks client update dispersion via parameter variance:

$$\mathrm{Var}(W_{t,i}) = \frac{1}{|S_t|} \sum_{k \in S_t} \|\Delta_{t,i}^{(k)}\|_2^2 - \|g_{t,i}\|_2^2$$

  Low variance = aligned progress; high variance = client drift.
- **Event-Driven Halting**: Synchronize when $\mathrm{Var}(W) > \theta_t$.
- **Midpoint Threshold Scaling**: $\theta_{t+1} = \mathrm{ar}_t \cdot \frac{\tau}{2s_t}$.

Research Gap in FDA-Opt

- FDA-Opt operated strictly on **full model parameters** (approximated via AMS sketches).
- Spatial compression (LoRA) and temporal adaptivity (FDA) had never been synthesized.
- Low-rank adapters exhibit high variance volatility requiring decoupled mechanics.

Background Knowledge : Synthesis of Foundational Precursors

Synthesizing RoLoRA (Spatial) and FDA-Opt (Temporal)

Dimension	**RoLoRA** (NeurIPS '25)	**FDA-Opt** (2025)	**FDA-RoLoRA** (**This Thesis**)
Optimization Focus	Spatial parameter volume	Temporal synchronization	**Joint Spatial + Temporal**
Parameter Scope	Low-rank adapters (A, B)	Full model weights	**Low-rank adapters (A, B)**
Aggregation	Alternating coordinates	Standard FedAvg / FedOpt	**Alternating coordinates**
Local Scheduling	Static fixed (E epochs)	Dynamic (Model variance)	**Dynamic (Subspace variance)**
Variance Tracking	None	Full-weight (AMS sketch)	**Exact Subspace** (No sketch)
Threshold Logic	None	Single scalar threshold	**Decoupled Dual-Track** (θ_A, θ_B)

Complementary Synergy

FDA-RoLoRA solves the core limitations of both parent papers: RoLoRA provides the strict aggregation linearity needed for stable variance tracking, while FDA-Opt provides the event-driven cadence needed to eliminate static schedule waste.

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Methodology : Dynamic Averaging in Federated LoRA

Taxonomy of Integration Avenues

Different PEFT architectures have distinct mathematical structures and optimization geometries; dynamic averaging cannot use a single one-size-fits-all rule.

- **FDA + FedAvg-LoRA**: Dynamic averaging with simultaneous (A, B) training.
  - *Drawback*: Inherits bilinear aggregation error; non-linear distortion destabilizes variance monitoring.
- **FDA + FFA-LoRA**: Dynamic averaging with permanently frozen A *[Future Work]*.
- **FDA + SLoRA**: Dynamic thresholding over sparse adapter coordinates *[Future Work]*.
- **FDA-RoLoRA**: **This Thesis**
  - Synthesizes **RoLoRA** (Chen et al., 2025) and **FDA-Opt** (Theologitis et al., 2025).
  - Alternating coordinates provide strict linearity $\rightarrow$ perfect substrate for variance tracking.
  - Maintains independent persistent variance thresholds (θ_A, θ_B) for each subspace.

Methodology : FDA-RoLoRA Architecture

Decoupled Subspace Variance Mechanics

RoLoRA isolates the optimization trajectories of down-projection (A) and up-projection (B). FDA-RoLoRA leverages this to isolate variance tracking and threshold evolution.

- **Active Subspace Mapping**: Server maps active threshold based on designated matrix $\mathcal{M}^{(r)} \in \{A, B\}$:

$$\theta_{\text{current}}^{(r)} = \begin{cases} \theta_A & \text{if } \mathcal{M}^{(r)} = A \\ \theta_B & \text{if } \mathcal{M}^{(r)} = B \end{cases}$$

- **Direct Subspace Variance**: Evaluated exclusively on low-rank coordinate tensors:

$$\text{Var}_{\text{subspace}}(\Delta^{(e)}) = \frac{1}{|S_r|} \sum_{i \in S_r} \|\Delta_i^{(e)}\|_2^2 - \left\| \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e)} \right\|_2^2$$

- Eliminates memory/compute overhead of mapping drifts back to the full weight space.

Methodology : Independent Halting & Local Training Bounds

Local Epoch Bounds ($e_{\min} = 1$)

Client training bounded in $[e_{\min}, e_{\max}]$ with $e_{\min} = 1$ standardized everywhere to guarantee baseline progress:

$$\text{Halt if } \text{Var}_{\text{subspace}}(\Delta^{(e)}) > \theta^{(r)}_{\text{current}} \ \vee \ e \geq e_{\max}$$

Evaluated at epoch boundaries $e \geq e_{\min}$.

Threshold Controller ($k \cdot e_{\max}$)

Scales threshold toward target horizon $k \cdot e_{\max}$:

$$\theta^{(r)}_{\text{target}} = \left(\frac{k \cdot e_{\max}}{e^*} \right) \text{Var}_{\text{subspace}}(\Delta^{(e^*)})$$

- **Default ($k = 0.5$)**: FDA-Opt midpoint ($e_{\max}/2$).
- **Aggressive ($k = 1.0$)**: Full ceiling target ($e_{\max}$).

Controller Dynamics: Non-IID Success vs. IID Midpoint Lag

Non-IID: Default $k = 0.5$ works almost perfectly containing drift and preventing collapse.
IID: $k = 0.5$ fails to reach the midpoint ($e_{\max}/2$); $k = 1.0$ pushes epochs higher (e.g. toward 20 under $e_{\max} = 40$), but overall struggles against static schedules.

Methodology : Threshold Evolution Mechanics & Controller Design

Primary Operational Controller: Standard Midpoint Scaling

Direct assignment without memory lag: $\theta_{\mathcal{M}^{(r)}}^{(\text{next})} = \theta_{\text{target}}^{(r)}$.

- **Parameter-Free**: No sensitivity to tuning hyperparameters.
- **Adaptive Response**: Responds instantly to changes in gradient drift variance.

Exploratory Temporal Smoothing Variants (Prototyped)

- **Simple Moving Average (SMA)**: Mean over past K active rounds: $\frac{1}{K}\sum_{k=0}^{K-1}\theta_{\text{target}}^{(r-k)}$.
- **Exponential Moving Average (EMA)**: Historical weighting: $\rho\theta_{\text{current}} + (1-\rho)\theta_{\text{target}}$.
- **Finding**: Prototyping revealed smoothing added threshold inertia/lag under distribution shifts without tangible accuracy benefit; hence retained as future work.

Methodology : Coordinate-Wise FedAvg & Protocol Flow

Coordinate-Wise FedAvg Aggregation

- Client drift: $\Delta_i^{(e^*)} = W_{i,e^*}^{(r)} - W_{\text{act}}^{(r)}$
- Cohort average: $\bar{\Delta}^{(r)} = \frac{1}{|S_r|} \sum_{i \in S_r} \Delta_i^{(e^*)}$
- Linear FedAvg update ($\eta_s = 1.0$):

$$W_{\text{act}}^{(r+1)} = W_{\text{act}}^{(r)} + \bar{\Delta}^{(r)}$$

- **Linear Aggregation**: Standard FedAvg avoids momentum artifacts, isolating gains strictly to dynamic variance tracking.

Protocol Execution Lifecycle

1. **Toggle**: Server selects active track A or B.
2. **Broadcast**: Sends $W_{\text{act}}^{(r)}$ and $\theta_{\text{current}}^{(r)}$ to S_r.
3. **Train**: Clients run local SGD steps ($\eta_c = 0.1$).
4. **Check**: Evaluate subspace variance at $e \geq e_{\min}$.
5. **Halt**: Return $\Delta_i^{(e^*)}$ when Var $> \theta$ or $e = e_{\max}$.
6. **Aggregate**: Server averages client drifts ($\bar{\Delta}^{(r)}$).
7. **Evolve**: Update active threshold $\theta_{\mathcal{M}^{(r)}}^{(\text{next})}$.
8. **Step**: Server applies FedAvg to active coordinate.

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Implementation : Modular 3-Tier Software Architecture

1. Server Orchestrator

- Global A, B matrices.
- Tracks (θ_A, θ_B) state.
- Coordinates RoLoRA toggle.
- Coordinate-wise FedAvg.

2. Client Nodes

- Private data partitions.
- Local SGD training loops.
- Subspace drift computation.
- Variance checks at $e \geq e_{\min}$.

3. Comm. Interface

- Low-rank serialization.
- PyTorch, HF & PEFT.
- Lightweight FDA-Opt core.

Key Engineering Decisions

- **FDA-Opt Base vs. FederatedScope-LLM**: The RoLoRA paper (Chen et al., NeurIPS 2025) used FederatedScope-LLM without a public release. We adopted and extended the open-source **FDA-Opt** repository (Theologitis et al., 2025), integrating HuggingFace PEFT and RoLoRA coordinate alternation into its modular PyTorch core.
- **Exact Variance over AMS Sketch**: Full-parameter FDA-Opt required randomized AMS sketching to bound memory. In low-rank coordinates ($r = 8$), parameters are so compact that exact subspace variance is computed with negligible overhead, eliminating sketching approximations.

Implementation : Foundation Model Setup & Data Partitioning

Foundation Model Backbone

- **Model**: `roberta-base` (125M params, 12 layers, 768 hidden; selected for multi-seed sweeps under hardware budget).
- **Adapter Placement**: Low-rank adapters injected into Query (W_q) and Value (W_v) projections across top encoder layers.
- **Adapter Configuration**: Rank $r = 4$, scaling factor $\alpha_{\text{lora}} = 8$ (0.47% trainable parameter share).
- **Spatial Reduction**: Backbone frozen; communicating only compact A, B factors cuts parameter transfer by $> 99\%$.

Resource Constraints & Simulation Setup

- **Dirichlet Partitioning**: Client data sampled via $p_k \sim \text{Dir}(\alpha)$ ($\alpha \in \{0.5, 1000.0\}$).
- **25k Dataset Cap**: Datasets capped at $\leq 25{,}000$ samples (MAX_SAMPLES $= 25{,}000$) due to compute limits.
- **Batch Sizes**: $B = 16$ (MNLI, QNLI, QQP), $B = 32$ (SST-2, following Chen et al., 2025).
- **Simulation Early Stopping**: Accuracy-gated early stopping to prevent redundant GPU-hours.
- **Benchmark Tasks**: GLUE suite (SST-2, QQP, QNLI, MNLI).

Implementation : Training Pipeline & Dedicated Testbed Deployment

Training Pipeline Mechanics

- **Alternation**: Track B on even rounds ($t = 0, 2, \dots$), Track A on odd rounds ($t = 1, 3, \dots$).
- **Exact Subspace Variance**: Evaluated centrally at $e \geq e_{\text{min}}$ without high-dimensional projection.
- **Optimization**: Client SGD ($\eta_c = 0.1$), server SGD step ($\eta_s = 1.0$) replicating FedAvg over active coordinate.
- **Orchestration**: Slurm cluster jobs supported; multi-process client worker orchestration.

Polytechnix Server Testbed

- **Server**: SoftNet Lab Polytechnix GPU server.
- **GPUs**: 2× NVIDIA Ampere A10 (24 GB VRAM each).
- **CPUs**: 2× Intel Xeon Silver 4310 (24c/48t, 2.10 GHz).
- **RAM**: 256 GB DDR4 system memory.
- **Comparability**: All 173 runs evaluated exclusively on this uniform platform to ensure perfect comparability.

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Evaluation : Experimental Setup & Benchmark Protocol

Model & Low-Rank Parameterization

- **Backbone**: `roberta-base` (125M params, 12 layers).
- **LoRA Setup**: Rank $r = 4$, $\alpha = 8$ on attention W_q, W_v (0.47% trainable parameters).
- **Uplink Payload**: 240 KB/client/round (sending only active factor A or B).
- **Batch Sizes**: $B = 16$ (MNLI, QNLI, QQP), $B = 32$ (SST-2, following Chen et al., 2025).
- **Optimization**: Mini-batch SGD ($\eta_c = 0.1$, $\eta_s = 1.0$).

Benchmark Matrix & Regimes

- **Tasks**: GLUE suite (MNLI, QNLI, QQP, SST-2) capped at 25,000 training samples.
- **Cohorts**: $C \in \{20, 50\}$ clients (full participation).
- **Regimes**: Homogeneous ($\alpha = 1000.0$) vs. Dirichlet Label Skew ($\alpha = 0.5$).
- **Baselines**: Fixed RoLoRA $E \in \{5, 10, 15, 20\}$.
- **FDA-RoLoRA**: Standardized $e_{\text{min}} = 1$ everywhere; $E_{\text{max}} \in \{20, 40\}$; controller $k \in \{0.5, 1.0\}$.
- **Scale**: **173 completed runs** across 86 configs.

Controller Horizon Dynamics: Default ($k = 0.5$) vs Aggressive ($k = 1.0$)

FDA-Opt default $k = 0.5$ targets midpoint ($E_{\text{max}}/2$). In Non-IID, it worked near-perfectly to halt drift. In IID, $k = 0.5$ lagged behind midpoint (e.g., failed to reach $\bar{e} = 20$ for $E_{\text{max}} = 40$); $k = 1.0$ pushed epochs close to 20, but overall struggled against static baselines.

Evaluation : Executive Performance Scorecard

Comprehensive Synthesis Across GLUE Benchmarks on `roberta-base` ($e_{\min} = 1$)

Task & Cohort	Regime	Top FDA-RoLoRA ($e_{\min} = 1$)	FDA Peak Acc. (%)	Top RoLoRA Baseline	RoLoRA Peak (%)	Δ Acc.
MNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**76.86 ± 0.07**	RoLoRA ($E = 10$)	76.60 ± 0.60	**+0.26%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**74.47 ± 0.08**	RoLoRA ($E = 10$)	74.24 ± 0.09	**+0.23%**
MNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**69.85 ± 2.38**	RoLoRA ($E = 10$)	65.85 ± 2.02	**+4.00%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 0.5$)	55.56 ± 1.17	RoLoRA ($E = 10$)	**56.55 ± 4.90**	-0.99%
QNLI ($C = 20$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**83.14 ± 0.24**	RoLoRA ($E = 10$)	81.83 ± 0.65	**+1.31%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**84.22 ± 0.29**	RoLoRA ($E = 10$)	83.50 ± 1.18	**+0.72%**
QNLI ($C = 50$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	82.77 ± 1.51	RoLoRA ($E = 20$)	**88.14 ± 0.00**	-5.37%
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 0.5$)	**75.69 ± 1.17**	RoLoRA ($E = 10$)	74.61 ± 7.18	**+1.08%**
QQP ($C = 20$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	**78.57 ± 2.99**	RoLoRA ($E = 10$)	75.47 ± 0.84	**+3.10%**
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**77.88 ± 0.46**	RoLoRA ($E = 10$)	77.69 ± 1.54	**+0.19%**
SST-2 ($C = 50$)	IID ($\alpha = 1000$)	FDA ($E_{\max} = 20, k = 1.0$)	89.68 ± 0.80	RoLoRA ($E = 20$)	**92.01 ± 0.46**	-2.33%
	Non-IID ($\alpha = 0.5$)	FDA ($E_{\max} = 20, k = 1.0$)	**85.95 ± 3.96**	RoLoRA ($E = 10$)	59.00 ± 7.86	**+26.95%**

Key Empirical Breakthroughs

Anti-Collapse on SST-2 Non-IID: RoLoRA crashes to 59.00%; FDA secures **85.95%** (+**26.95%** recovery).
Consensus Acceleration on MNLI: Breaks static plateau at 65.85% to **69.85%** (+**4.00%** outperformance).
Communication Reductions: Converges with up to **56.6% lower network bandwidth** (QNLI Non-IID $C = 50$).

Evaluation : Homogeneous Consensus (IID, $\alpha = 1000.0$)

The Compute Accelerator Mechanism in Shared-Gradient Environments

Optimization Dynamics & Controller Behavior

- **Consensus**: Updates align along shared trajectory ($\bar{\nu} \leq 1.0$).
- **Default ($k = 0.5$)**: FDA-Opt $E_{\max}/2$ lagged below midpoint ($E_{\max} = 40$ failed to reach $\bar{e} = 20$).
- **Aggressive ($k = 1.0$)**: Pushed epochs higher ($\bar{e} \to 20$), but struggled overall against static baselines.

Empirical Highlights

- **QQP ($C = 50$)**: Early stops in **14.0 rd** (164 MB) vs 30 rd (352 MB) → **53.3% bandwidth cut**.
- **MNLI ($C = 50$)**: Reaches **69.85**% vs 65.85% plateau (+**4.00**%).
- **QNLI ($C = 20$)**: Peak accuracy **83.14**% vs 81.83%.

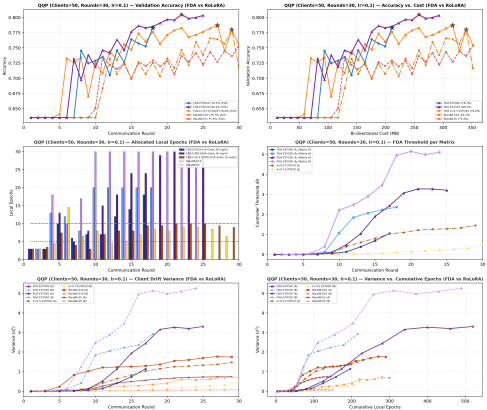


Figure: Multi-panel telemetry on QQP ($C = 50$, IID): (a) Acc/Rounds; (b) Acc/MB; (c) Epochs; (d) θ; (e) $\bar{\nu}$; (f) Var/Epochs.

Evaluation : Statistical Heterogeneity (Non-IID, $\alpha = 0.5$)

The Safety Brake Mechanism & Anti-Collapse Protection Under Label Skew

The "Safety Brake": Near-Flawless Performance

- Under Non-IID, FDA-Opt's controller worked **almost perfectly**.
- Dirichlet skew ($\alpha = 0.5$) causes drift variance to surge **4.0**×.
- Senses spikes and **throttles local epochs**, preventing catastrophic collapse (+26.95% SST-2).

Empirical Highlights

- **SST-2 (**$C = 50$**)**: RoLoRA crashes to 59.00%; FDA reaches **85.95**% (+**26.95**% recovery).
- **QNLI (**$C = 20$**)**: Suite-leading **84.22**% peak accuracy.
- **QNLI (**$C = 50$**)**: Early stops in 13.0 rd (152.3 MB) vs. 30.0 rd → **56.6**% bandwidth savings.

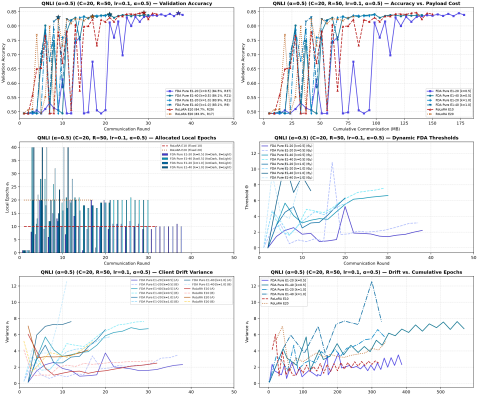


Figure: Multi-panel telemetry on QNLI under Dirichlet skew ($\alpha = 0.5$, $C = 20$): closed-loop tracking securing 84.22% accuracy.

Evaluation : Factor Asymmetry & Decoupled Tracking

Down-Projection (A) vs. Up-Projection (B) Optimization Geometry

Mathematical Factor Asymmetry

- **Down-Projection** $A \in \mathbb{R}^{d \times r}$: Compresses input features into low-rank bottleneck; Gaussian initialization.
- **Up-Projection** $B \in \mathbb{R}^{r \times d}$: Expands bottleneck representations back to hidden dimension; zero initialization.
- Exhibit distinct gradient magnitudes and variance volatility.

Decoupled Threshold Isolation

Maintaining independent thresholds (θ_A, θ_B) ensures that the inactive track remains frozen, eliminating cross-phase geometric interference.

Task & Cohort	Config	Mean $\bar{e}$	$\bar{e}_A$	$\bar{e}_B$
MNLI ($C = 20$, Non-IID)	$E_{\max} = 20, k = 1.0$	17.49	17.19	**17.76**
MNLI ($C = 50$, Non-IID)	$E_{\max} = 20, k = 0.5$	10.13	9.47	**10.80**
QNLI ($C = 20$, Non-IID)	$E_{\max} = 20, k = 1.0$	16.73	16.63	**16.81**
QNLI ($C = 50$, Non-IID)	$E_{\max} = 20, k = 0.5$	10.48	9.83	**11.13**
QQP ($C = 50$, Non-IID)	$E_{\max} = 20, k = 1.0$	14.26	13.83	**14.63**

Key Finding

Up-projection factor B consistently absorbs deeper local epoch allocations ($\bar{e}_B > \bar{e}_A$) across benchmarks, confirming that low-rank factors require decoupled dynamic regulation.

Evaluation : Communication Efficiency: Milestone Gates

Theologitis-Style Analysis: Rounds and Total Payload to Target Accuracies

Task ($C = 20$)	Regime	Methodology	Milestone Gate 1	Milestone Gate 2	Milestone Gate 3	Total MB
SST-2	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@80%: 23 rd (107.8 MB)	R@85%: 24 rd (112.5 MB)	R@88%: 24 rd (112.5 MB)	117.2 MB
		FDA-RoLoRA ($E_{\max} = 20, k = 1.0$)	**R@80%: 8 rd (37.5 MB)**	**R@85%: 8 rd (37.5 MB)**	**R@88%: 9 rd (42.2 MB)**	**65.6** MB
MNLI	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@65%: 28 rd (131.2 MB)	R@72%: 43 rd (201.6 MB)	R@74%: 50 rd (234.4 MB)	234.4 MB
		FDA-RoLoRA ($E_{\max} = 20, k = 1.0$)	**R@65%: 19 rd (89.1 MB)**	**R@72%: 25 rd (117.2 MB)**	**R@74%: 36 rd (168.8 MB)**	**178.1** MB
QNLI	Non-IID ($\alpha = 0.5$)	RoLoRA ($E = 10$)	R@80%: 17 rd (79.7 MB)	R@83%: 25 rd (117.2 MB)	R@84%: — (Unattained)	154.7 MB
		FDA-RoLoRA ($E_{\max} = 20, k = 1.0$)	**R@80%: 15 rd (70.3 MB)**	**R@83%: 21 rd (98.4 MB)**	**R@84%: 25 rd (117.2 MB)**	**124.2** MB

2.87× Convergence Speedup (SST-2)

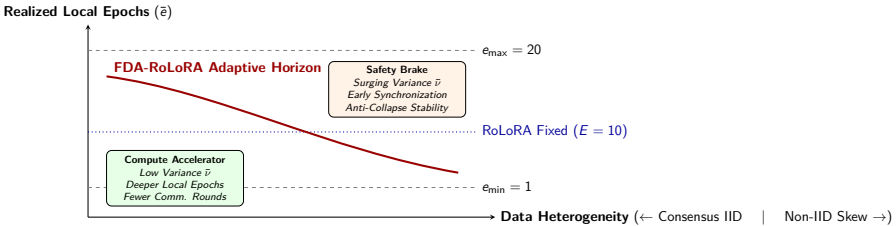
Crosses 80% in **8 rounds** (37.5 MB) vs. 23 rounds (107.8 MB) → **65.2% bandwidth savings**.

Higher Generalization Ceilings (QNLI)

Reaches 84% gate in **25 rounds** (117.2 MB), whereas static RoLoRA fails to ever cross 84%.

Evaluation : Cross-Regime Synthesis: The Dual Frontier

Autonomous Operating Modes Governed by Coordinate Variance



Consensus Accelerator: Low drift → scales $\theta \to \bar{e} \to e_{max}$, slashing round latency.

Anti-Drift Safety Brake: High drift → throttles $\bar{e} \to e_{min}$, preventing catastrophic collapse.

Evaluation : Practical Deployment Guidelines

Four Core Architectural Principles for Real-World Federated Systems

1. Universal Base Progress ($e_{\min} = 1$)

Standardize $e_{\min} = 1$ universally everywhere to ensure baseline progress (exploratory runs testing $e_{\min} = 3$ showed negligible difference).

2. Controller Dynamics in IID vs. Non-IID

- **Non-IID ($k = 0.5$)**: FDA-Opt default ($E_{\max}/2$) works near-perfectly to contain drift and prevent collapse.
- **IID ($k = 1.0$ vs 0.5)**: $k = 0.5$ lags below midpoint; $k = 1.0$ pushes epochs higher (toward 20 for $E_{\max} = 40$), but struggles against static schedules.

3. Decouple Factor Thresholds

Always maintain independent (θ_A, θ_B) states. Freezing the inactive threshold respects low-rank factor asymmetry.

4. Calibrate Early Stopping Under Skew

In non-IID regimes, configure wider smoothing windows ($W \geq 5$) and a round floor ($R_{\min} \geq 0.6 \times R_{\max}$) to tolerate alternating validation oscillations.

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Conclusion : Synthesis of Scientific Contributions

- **Bridging RoLoRA & FDA-Opt**: Unified alternating low-rank adaptation (**RoLoRA**, Chen et al., NeurIPS 2025) with dynamic variance-driven averaging (**FDA-Opt**, Theologitis et al., 2025) into a coherent dual-axis federated framework.
- **Mathematical Consistency via Alternating Isolation**: Alternating coordinate optimization solves the bilinear aggregation inconsistency, providing strictly linear updates that enable stable, distortion-free geometric variance monitoring.
- **Decoupled Subspace Geometric Monitoring**: Evaluating drift variance directly on isolated low-rank coordinate tensors eliminates the memory and communication overhead of full-model gradient approximations.
- **Dual Frontier Operational Model**: Empirically demonstrated across 173 runs on `roberta-base`: acts as a *Compute Accelerator* in consensus (up to 56.6% bandwidth savings) and a *Safety Brake* under skew (+**26.95**% anti-collapse stability recovery on SST-2).
- **Full Empirical Foundation Benchmark**: Completed comprehensive multi-seed benchmark across GLUE tasks on dedicated Polytechnix GPU hardware.

Conclusion : Limitations & Future Research Horizons

Limitations

- **Deterministic Alternation**: Alternates A and B in strict sequence; asymmetric or adaptive switching cadences remain unexplored.
- **Simulated Multi-Process**: Executed on a dedicated multi-GPU server; physical wide-area latency and client dropout left unmodeled.
- **Backbone Scope**: Evaluated on `roberta-base` (125M); multi-billion parameter LLMs (e.g., Llama-3 8B) remain future targets.

Future Horizons

- **Advanced Controllers**: PID-regulated threshold scaling and variance-acceleration damping.
- **Broader PEFT Variants**: Formalizing FDA with FFA-LoRA, SLoRA, and dynamic rank allocation (AdaLoRA).
- **Quantization & Edge**: Synthesizing with 4-bit QLoRA for constrained edge devices.
- **Physical Edge Testbed**: Deployment across heterogeneous physical IoT and mobile hardware.

Key References

S. Chen, Y. Guo, Y. Ju, H. Dalal, Z. Zhu, and A. Khisti.
Robust Federated Finetuning of LLMs via Alternating Optimization of LoRA.
In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

M. Theologitis, V. Samoladas, and A. Deligiannakis.
FDA-Opt: Communication-Efficient Federated Fine-Tuning of Language Models.
arXiv preprint arXiv:2505.04535, 2025.

E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen.
LoRA: Low-Rank Adaptation of Large Language Models.
In *International Conference on Learning Representations (ICLR)*, 2022.

B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas.
Communication-Efficient Learning of Deep Networks from Decentralized Data.
In *AISTATS*, pages 1273–1282, 2017.

I. Sharfman, A. Schuster, and D. Keren.
A Geometric Approach to Monitoring Threshold Functions over Distributed Data Streams.
In *ACM SIGMOD*, pages 301–312, 2006.

Thank You!

Questions & Discussion

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Open-Source Repository: `https://github.com/Asterinos1/FDA-RoLoRA`